YEAR 10 PHYSICS, AQA GCSE, UNIT REVIEW

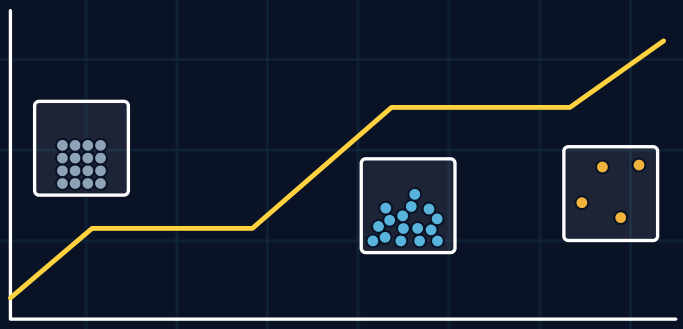
# PARTICLE MODEL OF MATTER

## UNIT REVIEW

AFTER LESSON 8

Mixed exam practice from the whole unit, a fix-it log, the equation toolkit, and extension pages on gas pressure and volume for GCSE Physics (Triple Science) students.

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

REFERENCE

# EQUATION TOOLKIT

Every equation in the unit. AQA prints all of them on the equation sheet in the exam, but you still need to know what each symbol means, its unit, and how to use it.

1. Cover the quantity you want to find.
2. What is left side by side? **Multiply**.
3. One above the other? **Divide** the top by the bottom.

Example: to find $V$ from $\rho = \frac{m}{V}$, cover $V$. $m$ is above $\rho$, so $V = \frac{m}{\rho}$.

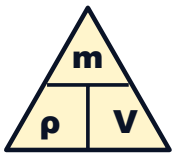
## $\rho = \frac{m}{V}$

Lesson 3

**density = mass ÷ volume**

kg/m³, kg, m³ (or g/cm³, g, cm³)

$m = \rho V$ and $V = \frac{m}{\rho}$



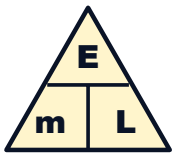

## $E = m L$

Lesson 6

**energy for a change of state = mass × specific latent heat**

J, kg, J/kg

$L = \frac{E}{m}$ and $m = \frac{E}{L}$



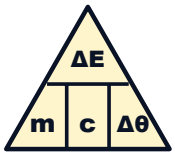

## $\Delta E = m c \Delta\theta$

Lesson 7

**change in thermal energy = mass × specific heat capacity × temperature change**

J, kg, J/kg °C, °C

$c = \frac{\Delta E}{m \Delta\theta}$ and $\Delta\theta = \frac{\Delta E}{m c}$




## $E = V I t$

Lesson 8

**energy supplied = potential difference × current × time**

J, V, A, s

From $P = V I$ and $E = P t$ (Electric Circuits)

## $p V = \text{constant}$

Triple only

**pressure × volume = constant, for a fixed mass of gas at constant temperature**

Pa, m³

Use it as $p_1 V_1 = p_2 V_2$

### 1

Write what you know. Convert first: g → kg, cm³ → m³, kJ → J, min → s.

### 2

Write the equation in symbols.

### 3

Rearrange if you need to, then substitute the numbers.

### 4

Give the answer with a unit. Ask: is it sensible?

AFTER LESSON 8, AT HOME OR IN CLASS

# UNIT REVIEW

Mixed questions from the whole unit, like a real exam. Try each question without looking back. The equation toolkit is on the page before this one. Then mark your answers with the answer edition and fill in the “fix it” log.

AQA-STYLE

## MIXED EXAM PRACTICE

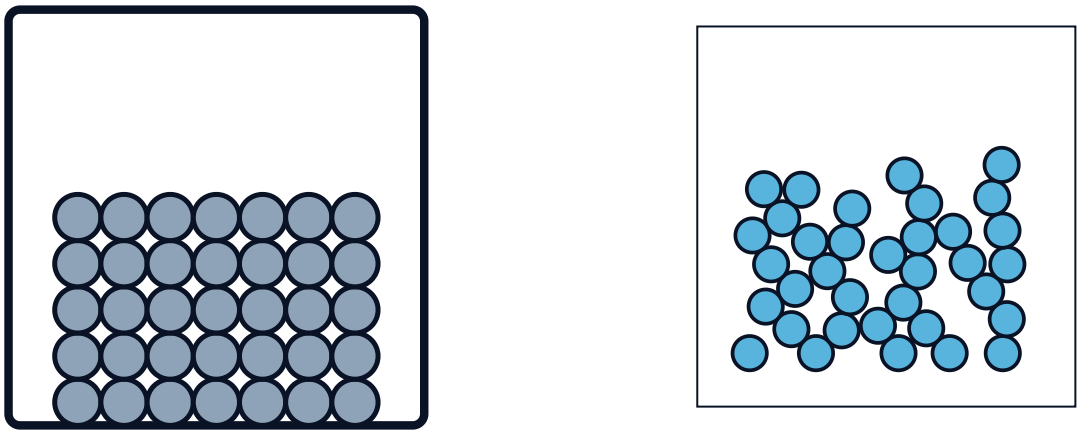
39 marks, about 45 minutes

**1** Draw **one** line from each state of matter to the description of its particles. **[3 marks]**

| | |
|---|---|
| solid | far apart, moving quickly in all directions |
| liquid | close together in a regular pattern, vibrating |
| gas | close together, moving past each other |

**All three correct (3); two correct (2); one correct (1).**

**2** The box shows the particles in a solid. In the empty box, draw the particles of the same substance as a **liquid**. **[2 marks]**



**Particles touching (1); in a random arrangement, about as many as in the solid, the same size (1).**

**3** A sealed flask contains 200 g of ice. The ice melts. **[2 marks]**

(a) State the mass of the water in the flask.

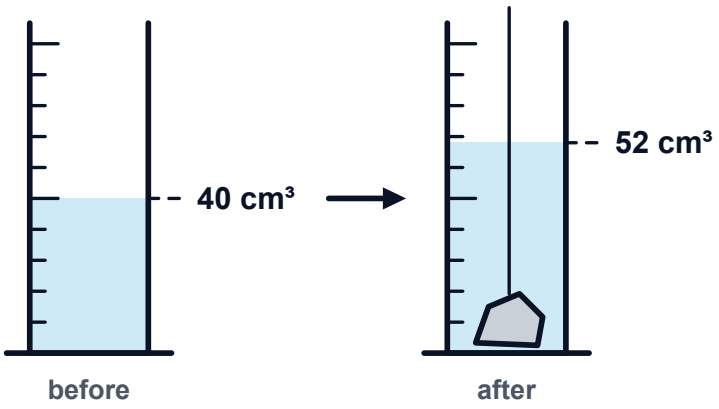
(b) Explain your answer.

**(a) 200 g (1). (b) The number of particles does not change: none are added or removed (1).**

**4** The density of aluminium is 2700 kg/m³. Calculate the mass of an aluminium block of volume 0.040 m³. **[3 marks]**

| | |
|---|---|
| EQUATION | $\rho = \frac{m}{V}$ |
| REARRANGE | $m = \rho V$ (1) |
| SUBSTITUTE | $m = 2700 \times 0.040$ (1) |
| ANSWER | **108 kg (1)** unit: |

**5** A student measures the density of a metal bolt. The bolt has a mass of 93.6 g. The student lowers it into a measuring cylinder of water. **[5 marks]**




(a) Calculate the density of the bolt in g/cm³.

| | |
|---|---|
| VOLUME | $V = 52 - 40 = 12$ cm³ (1) |
| EQUATION | $\rho = \frac{m}{V}$ |
| SUBSTITUTE | $\rho = \frac{93.6}{12}$ (1) |
| ANSWER | **7.8 g/cm³ (1)** unit: |

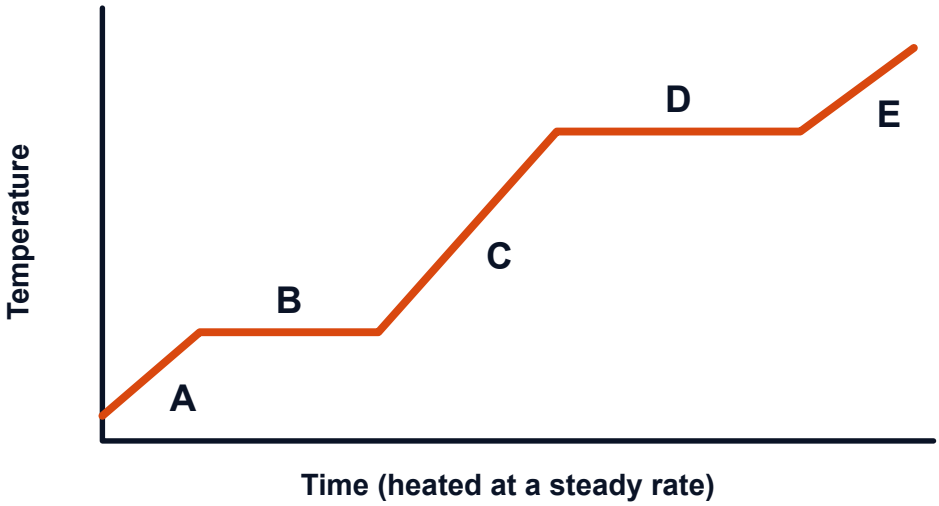
(b) The student reads the top of the meniscus both times instead of the bottom. Explain how this affects the volume of the bolt that the student calculates.

**It makes no difference (1): both readings are too high by the same amount, so their difference is unchanged (1).**

**6** A sealed metal can of air is heated. Explain, in terms of particles, what happens to the pressure of the air. **[3 marks]**

**The molecules move faster (their average kinetic energy increases) (1). They collide with the walls more often (1) and with more force, so the pressure increases (1).**

**7** A solid is heated at a steady rate. The graph shows how its temperature changes. **[4 marks]**




(a) Which section shows the substance boiling?

| A | B | C | D | E |
|---|---|---|---|---|

(b) Explain why the temperature does not change during section B, even though the substance is still being heated.

**The substance is melting (1). The energy supplied increases the potential energy of the particles (1), not their kinetic energy (so the temperature stays constant) (1).**

**8** State what is meant by the internal energy of a system. **[2 marks]**

**The total kinetic energy and potential energy (1) of all the particles in the system (1).**

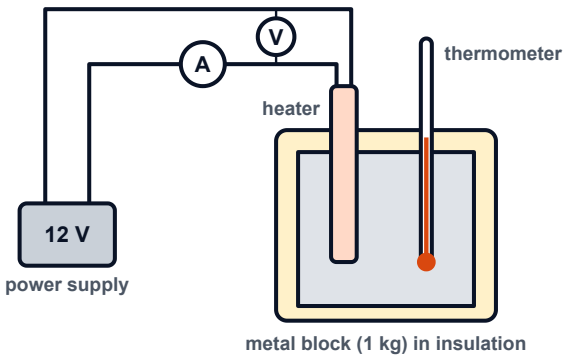
**9** Calculate the energy needed to melt 0.25 kg of ice at 0 °C. Specific latent heat of fusion of water = 334 000 J/kg **[2 marks]**

| | |
|---|---|
| EQUATION | $E = m L$ |
| SUBSTITUTE | $E = 0.25 \times 334\,000$ (1) |
| ANSWER | **83 500 J (1)** unit: |

**10** 500 g of water is heated from 20 °C to 60 °C. Calculate the energy transferred to the water. Specific heat capacity of water = 4200 J/kg °C **[4 marks]**

| | |
|---|---|
| CONVERT | **m = 0.500 kg (1)** |
| TEMPERATURE CHANGE | $\Delta\theta = 60 - 20 = 40$ °C (1) |
| EQUATION | $\Delta E = m c \Delta\theta$ |
| SUBSTITUTE | $\Delta E = 0.500 \times 4200 \times 40$ (1) |
| ANSWER | **84 000 J (1)** unit: |

**11** A student measures the specific heat capacity of a 1.0 kg iron block with an immersion heater. **[7 marks]**




(a) The heater runs at 12 V and 3.0 A for 300 s. Calculate the energy supplied. Use $E = V I t$

| | |
|---|---|
| EQUATION | $E = V I t$ |
| SUBSTITUTE | $E = 12 \times 3.0 \times 300$ (1) |
| ANSWER | **10 800 J (1)** unit: |

(b) The temperature of the block rises by 20 °C. Calculate the specific heat capacity of iron given by these results.

| | |
|---|---|
| EQUATION | $\Delta E = m\,c\,\Delta\theta$ |
| REARRANGE | $c = \dfrac{\Delta E}{m\,\Delta\theta}$ (1) |
| SUBSTITUTE | $c = \dfrac{10\,800}{1.0 \times 20}$ (1) |
| ANSWER | **540 J/kg °C (1)** unit: |

(c) The accepted value is 450 J/kg °C. Explain why the student's value is higher, and suggest one improvement.

**Some of the energy supplied was transferred to the surroundings, so the temperature rise was smaller than it should be (1). Insulate the block (1).**

12 Give one similarity and one difference between specific heat capacity and specific latent heat. **[2 marks]**

**Similarity: both are an amount of energy for 1 kg of a substance (1). Difference: specific heat capacity is for a temperature change of 1 °C; specific latent heat is for a change of state with no temperature change (1).**

REVIEW LATER

## FIX-IT LOG

Turn every lost mark into a target.

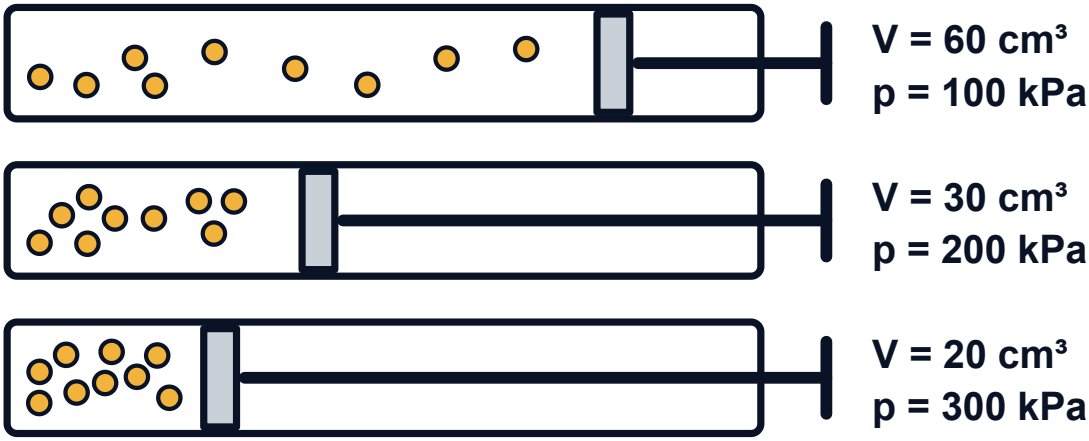
| Q | What went wrong? | The correct physics (or method) is… | Lesson to revisit |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

# PRESSURE AND VOLUME TRIPLE ONLY

Only for students taking GCSE Physics (Triple Science). Combined Science students can skip these pages.

A gas can be **compressed** (squashed) or allowed to **expand** by changing the pressure on it. The pressure of a gas produces a net force **at right angles** to the wall of its container, or to any surface in the gas.

## SQUASHING A GAS AT CONSTANT TEMPERATURE




The same trapped gas, squashed slowly so that its temperature stays the same

The temperature is constant, so the molecules move at the same average speed. In a **smaller** volume they are closer together, so they hit each square centimetre of wall **more often**: the pressure is bigger. In a **bigger** volume the molecules are more spread out and hit the walls less often: the pressure is smaller.

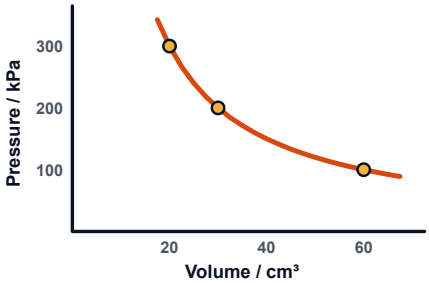
$$pV = \text{constant}$$

**pressure × volume = constant**

$p$ in Pa, $V$ in m³. For a fixed mass of gas at constant temperature.

Use it as $p_1 V_1 = p_2 V_2$: before and after. Any units will do if they are the same on both sides.

Halve the volume and the pressure doubles. Three times the volume gives a third of the pressure. Pressure is **inversely proportional** to volume, so a graph of pressure against volume is a curve.



**WITH YOUR TEACHER** SLIDE 49

**A gas has a volume of 60 cm³ at a pressure of 100 kPa. It is squashed to 20 cm³ at constant temperature. Calculate the new pressure.**

| | |
|---|---|
| EQUATION | $p_1 V_1 = p_2 V_2$ |
| REARRANGE | $p_2 = \frac{p_1 V_1}{V_2}$ |
| SUBSTITUTE | $p_2 = \frac{100 \times 60}{20}$ |
| ANSWER | $p_2 = 300$ kPa |

**YOUR TURN**

**A gas has a volume of 40 cm³ at a pressure of 120 kPa. It expands to 80 cm³ at constant temperature. Calculate the new pressure.**

| | |
|---|---|
| EQUATION | $p_1 V_1 = p_2 V_2$ |
| REARRANGE | $p_2 = \frac{p_1 V_1}{V_2}$ |
| SUBSTITUTE | $p_2 = \frac{120 \times 40}{80}$ |
| ANSWER | $p_2 = 60$ kPa |

PRACTISE

## CHECK IT

Answers are in the answer edition.

**1** Draw a ring around the correct answer in each box.

The force that a gas exerts on a wall acts

| | | |
|---|---|---|
| along the wall | **at right angles to the wall** | in a random direction |

A gas expands at constant temperature. Its pressure

| | | |
|---|---|---|
| increases | **decreases** | stays the same |

**2** A diver's air bubble has a volume of 12 cm³ where the pressure is 300 kPa. It rises to the surface, where the pressure is 100 kPa. Calculate its volume at the surface. Assume the temperature is constant. **[3 marks]**

| | |
|---|---|
| EQUATION | $p_1 V_1 = p_2 V_2$ |
| REARRANGE | $V_2 = \frac{p_1 V_1}{p_2}$ (1) |
| SUBSTITUTE | $V_2 = \frac{300 \times 12}{100}$ (1) |
| ANSWER | **36 cm³ (1)** unit: |

**3** Explain, in terms of particles, why the pressure of a gas decreases when its volume is increased at constant temperature. **[3 marks]**

**The molecules are more spread out (1), so they collide with the walls less often (1). The speed of the molecules has not changed (constant temperature), so the total force on each unit area of wall, the pressure, is smaller (1).**

TRIPLE SCIENCE ONLY (AQA PHYSICS 4.3.3.3)

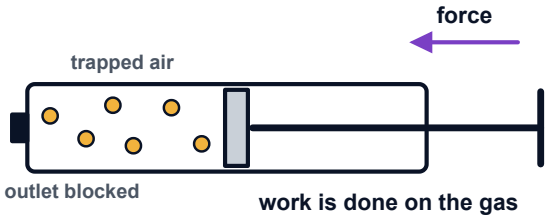
# DOING WORK ON A GAS TRIPLE ONLY

Only for students taking GCSE Physics (Triple Science).

> ! **Work** is the transfer of energy by a force. Doing work on a gas increases the **internal energy** of the gas, and can cause its temperature to increase.

Block the outlet of a bicycle pump and push the handle in quickly. The pump gets warm.

1. Your force moves the piston, so you do **work** on the trapped air.
2. Energy is transferred to the air: its **internal energy** increases.
3. The molecules bounce off the moving piston faster than they arrived, so their average kinetic energy increases.
4. The **temperature** of the air rises.




**Get the marks:** The air heats up when it is squashed **quickly**, because the energy has no time to escape. On the pages before this one, the gas was squashed slowly, so it stayed at the temperature of the room.

PRACTISE

## CHECK IT

Answers are in the answer edition.

**1** Complete the sentences.

Work is the transfer of ___energy___ **(energy / force / pressure)** by a ___force___ **(force / current / temperature)**.

When work is done on a gas, its internal energy ___increases___ **(increases / decreases / stays the same)**.

**2** A cyclist pumps up a tyre. Explain why the end of the pump becomes warm. **[3 marks]**

**The cyclist does work on the air in the pump (a force compresses it) (1). This transfers energy to the air, increasing its internal energy (1), so the temperature of the air (and then the pump) increases (1).**

CHECKPOINT

## QUICK CHECK

Slide 52, Triple Science pressure and volume. Answer each question, then check it as your teacher reveals the answer.

**1**

In which direction does the force of a gas on a wall act?

**At right angles to the wall.**

**2**

The volume of a gas is doubled at constant temperature. What happens to its pressure?

**It halves.**

**3**

A gas at 200 kPa and 0.50 m³ expands to 2.0 m³ at constant temperature. Calculate the new pressure.

$$p_2 = \frac{200 \times 0.50}{2.0} = 50$$

**kPa**

**4**

Name the two conditions needed for $pV = \text{constant}$.

**A fixed mass of gas at a constant temperature.**

**5**

What is work?

**The transfer of energy by a force.**

**6**

Work is done on a gas. What happens to its internal energy?

**It increases, so the gas can get hotter.**

REFERENCE

# KEY WORDS

Learn the meaning and the spelling. In exams, precise words earn marks: say “the particles vibrate about fixed positions”, not “the particles shake”.

**Particle model** L1: The idea that all matter is made of tiny particles, whose arrangement and motion explain how solids, liquids and gases behave

**Change of state** L1: Melting, freezing, boiling, evaporating, condensing or sublimating

**Sublimating** L1: Changing straight from a solid to a gas

**Physical change** L1: A change in which no new substance is made; the material recovers its original properties if the change is reversed

**Conserved** L1: Staying the same in total: mass is conserved in a change of state

**Random motion** L2: Movement in all directions, at a range of speeds

**Gas pressure** L2: The force on each square metre of a surface, caused by gas molecules colliding with it

**Density (ρ)** L3: The mass of each unit of volume, in kg/m³ or g/cm³

**Volume** L3: The amount of space something takes up, in m³ or cm³

**Displacement** L4: Finding the volume of an object from the volume of water it pushes out of the way

**Meniscus** L4: The curved surface of a liquid in a tube; read the bottom of it

**Resolution** L4: The smallest change in a quantity that an instrument can show

**Internal energy** L5: The total kinetic energy and potential energy of all the particles that make up a system

**Melting point** L5: The temperature at which a substance changes between solid and liquid

**Boiling point** L5: The temperature at which a substance boils, or condenses

**Latent heat** L6: The energy needed for a substance to change state, with no change in temperature

**Specific latent heat (L)** L6: The energy needed to change the state of 1 kg of a substance with no change in temperature, in J/kg

**Fusion** L6: Melting: the change from solid to liquid

**Vaporisation** L6: The change from liquid to vapour

**Specific heat capacity (c)** L7: The energy needed to raise the temperature of 1 kg of a substance by 1 °C, in J/kg °C

**Temperature change (Δθ)** L7: Final temperature minus starting temperature, in °C

**Joulemeter** L8: A meter that measures the energy supplied to an electrical appliance

**Insulation** L8: Material that reduces energy transfer to the surroundings

**Pascal (Pa)** Triple: The unit of pressure: 1 Pa = 1 N/m²

**Work** Triple: The transfer of energy by a force

FINISH

# PROGRESS TRACKER

Colour one circle for each statement: **red** = not yet, **amber** = getting there, **green** = I can do this without help. Revisit your reds first.

| I CAN… | R / A / G |
|---|---|
| **01: Solids, liquids and gases** | |
| draw and recognise particle diagrams for a solid, a liquid and a gas | 🔴 🟡 🟢 |
| describe the arrangement and motion of the particles in each state | 🔴 🟡 🟢 |
| name the changes of state | 🔴 🟡 🟢 |
| explain why mass is conserved in a change of state | 🔴 🟡 🟢 |
| explain why a change of state is a physical change | 🔴 🟡 🟢 |
| **02: Gas particles and pressure** | |
| describe the random motion of gas molecules | 🔴 🟡 🟢 |
| link temperature to the average kinetic energy of the molecules | 🔴 🟡 🟢 |
| explain how collisions with the walls cause pressure | 🔴 🟡 🟢 |
| explain why heating a gas at constant volume increases its pressure | 🔴 🟡 🟢 |
| **03: Density** | |
| calculate density with $\rho = \frac{m}{V}$ | 🔴 🟡 🟢 |
| use kg/m³ and g/cm³ and convert between them | 🔴 🟡 🟢 |
| rearrange the equation to find a mass or a volume | 🔴 🟡 🟢 |
| explain differences in density using the particle model | 🔴 🟡 🟢 |
| **04: Measuring density (required practical)** | |
| measure the density of a regular solid | 🔴 🟡 🟢 |
| use displacement to find the volume of an irregular solid | 🔴 🟡 🟢 |
| measure the density of a liquid | 🔴 🟡 🟢 |
| explain how to make the measurements accurate | 🔴 🟡 🟢 |
| describe the method in a 6-mark answer | 🔴 🟡 🟢 |
| **05: Internal energy and heating curves** | |
| state what internal energy is | 🔴 🟡 🟢 |
| describe what heating does to the particles | 🔴 🟡 🟢 |

| I CAN… | R / A / G |
|---|---|
| explain why the temperature is constant during a change of state | 🔴 🟡 🟢 |
| interpret heating and cooling graphs | 🔴 🟡 🟢 |
| **06: Specific latent heat** | |
| define specific latent heat | 🔴 🟡 🟢 |
| calculate energy, mass or $L$ with $E = m\,L$ | 🔴 🟡 🟢 |
| tell fusion from vaporisation | 🔴 🟡 🟢 |
| describe how to measure the specific latent heat of fusion of water | 🔴 🟡 🟢 |
| **07: Specific heat capacity** | |
| state what a temperature rise depends on | 🔴 🟡 🟢 |
| define specific heat capacity | 🔴 🟡 🟢 |
| calculate with $\Delta E = m\,c\,\Delta\theta$ | 🔴 🟡 🟢 |
| rearrange to find $c$ or $\Delta\theta$ | 🔴 🟡 🟢 |
| choose between specific heat capacity and specific latent heat | 🔴 🟡 🟢 |
| **08: Measuring specific heat capacity (required practical)** | |
| describe the method for measuring specific heat capacity | 🔴 🟡 🟢 |
| calculate the energy supplied with $E = V\,I\,t$ | 🔴 🟡 🟢 |
| calculate specific heat capacity from results | 🔴 🟡 🟢 |
| use the gradient of a temperature–energy graph | 🔴 🟡 🟢 |
| explain why the result is too high and how to improve it | 🔴 🟡 🟢 |